

The Program in Action

When you run *digitalclock.py*, the output will look like this:

```
|_| |_| * |_| |_| * |_| |_|
|_| |_| * |_| |_| * |_| |_|
```

Press Ctrl-C to quit.

How It Works

The digital clock program looks similar to Project 14, “Countdown.” Not only do they both import the *sevseg.py* module, but they must split up the multiline strings returned by *sevseg.getSevSegStr()* with the *splitlines()* method. This allows us to put a colon made of asterisks in between the digits for the hour, minute, and second sections of the clock. Compare this code with the code in Countdown to see how it is similar and how it is different.

```
1. """Digital Clock, by Al Sweigart al@inventwithpython.com
2. Displays a digital clock of the current time with a seven-segment
3. display. Press Ctrl-C to stop.
4. More info at https://en.wikipedia.org/wiki/Seven-segment\_display
5. Requires sevseg.py to be in the same folder.
6. View this code at https://nostarch.com/big-book-small-python-projects
7. Tags: tiny, artistic"""
8.
9. import sys, time
10. import sevseg # Imports our sevseg.py program.
11.
12. try:
13.     while True: # Main program loop.
14.         # Clear the screen by printing several newlines:
15.         print('\n' * 60)
16.
17.         # Get the current time from the computer's clock:
18.         currentTime = time.localtime()
19.         # % 12 so we use a 12-hour clock, not 24:
20.         hours = str(currentTime.tm_hour % 12)
21.         if hours == '0':
22.             hours = '12' # 12-hour clocks show 12:00, not 00:00.
23.         minutes = str(currentTime.tm_min)
24.         seconds = str(currentTime.tm_sec)
25.
26.         # Get the digit strings from the sevseg module:
27.         hDigits = sevseg.getSevSegStr(hours, 2)
28.         hTopRow, hMiddleRow, hBottomRow = hDigits.splitlines()
29.
30.         mDigits = sevseg.getSevSegStr(minutes, 2)
31.         mTopRow, mMiddleRow, mBottomRow = mDigits.splitlines()
32.
33.         sDigits = sevseg.getSevSegStr(seconds, 2)
```